

What is the topic of your project?

Our project examines gene expression data from cancer patients to determine whether the type of cancer a patient has can be identified based on which genes are active in the tumor. Using the “Gene Expression RNA-Seq” data set from the UCI Machine Learning Repository, which was collected from The Cancer Genome Atlas - Pan Cancer project. The dataset uses 5 different types of cancer: breast, colon, kidney, lung and prostate. Our goal is to determine whether the statistical methods can identify patterns in gene activity that correlate with specific cancers.

What key issues or problems do you plan to tackle?

The main problem is the dataset size. Using over 20,000 gene measurements with only 800 patient samples. The plan is to narrow our focus using a smaller, more manageable set of genes and using statistical tests to determine which of them differ most between cancer types.

What is your hypothesis statement?

First, we’d like to explore whether different tumor types have distinct gene-expression profiles. We hypothesize that these profiles should *differ* between tumor types. From here, we could potentially explore *which genes* are most informative for the tumor type. We could hypothesize that there is a *small subset* of genes that contain the *majority of information* needed to determine a tumor type.

Describe the dataset you plan to use, including the variables you will analyze, and provide an online link to the dataset. Detailed analysis plans will be included in the final written project.

Link to the dataset: <https://archive.ics.uci.edu/dataset/401/gene+expression+cancer+rna+seq>

Our selected dataset contains 801 samples, each belonging to one of five tumor types: breast cancer (BRCA), kidney renal clear cell carcinoma (KIRC), colon adenocarcinoma (COAD), lung adenocarcinoma (LUAD), and prostate adenocarcinoma (PRAD). Each sample is a molecular profile of

its tumor type, with over 20,000 gene measurements detailing how active each gene is. Though initially intimidating, the data is neatly organized and contained in just two CSV files, allowing us to use R packages and functions that we learned in class to preprocess and analyze effectively.

What questions or concerns do you have regarding your project?

- ❖ Which statistical methods would be most appropriate for comparing gene-expression levels across the 5 tumour types?
- ❖ How can we tell whether the differences we find between cancer types are statistically meaningful and are not just due to random variation?
- ❖ Are there certain genes that are especially useful for distinguishing one tumour from another?
- ❖ How should we deal with the very large number of gene measurements compared to the relatively small number of patient samples?
- ❖ If we find a small set of genes that separates the tumour types well, how can we test whether those genes are actually reliable and not just specific to this dataset?
- ❖ Are there any important limitations in using the dataset, such as differences in sample size between the 5 cancer types?
- ❖ How can we present the results in a way that clearly shows which genes or gene-expression patterns are most associated with each tumour type?